矢量分析基础 & 结论

$\vec{r} = (x, y, z)$

$\nabla \left(\frac{1}{r}\right) = -\frac{\vec{r}}{r^3}$

$\nabla (\nabla \times \vec{A}) = 0$

$\nabla \times (\nabla \cdot f) = 0$.

$\nabla \cdot (\varphi \vec{A}) = \nabla \varphi \cdot \vec{A} + \varphi \nabla \cdot \vec{A}$

$\nabla \times (\nabla \times \vec{A}) = \nabla (\nabla \cdot A) - \nabla^2 A$

$\delta(x) = \begin{cases} \infty, & x=0 \\ 0, & x \neq 0 \end{cases}$

$\int_{-\infty}^{+\infty} \delta(x)\, dx = 1$

$\int_{-\infty}^{+\infty} \delta(x) f(x)\, dx = f(0)$

$\nabla \left(\frac{\vec{r}}{r^3}\right) = 4\pi \delta(\vec{r}) = \nabla \left(\frac{\hat{r}}{r^2}\right)$

$\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right)$

$= \frac{\partial}{\partial x}\vec{i} + \frac{\partial}{\partial y}\vec{j} + \frac{\partial}{\partial z}\vec{k}$.

1.1 电荷与电场

1° E

$$\vec{F}=\frac{1}{4\pi\varepsilon_0}\cdot\frac{QQ'\cdot\vec{r}}{r^3}=Q'\vec{E}\Longrightarrow\vec{E}=\frac{1}{4\pi\varepsilon_0}\cdot\frac{Q\vec{r}}{r^3}.$$

$\Downarrow$ 连续电荷

$$\vec{E}=\frac{1}{4\pi\varepsilon_0}\int_V\frac{\rho\cdot\vec{r}}{r^3}dV\Rightarrow\frac{1}{4\pi\varepsilon_0}\int_V\frac{\rho(\vec{x}')\vec{r}}{r^3}dV'=E(\vec{x})$$

2° E的散度

$$\nabla\cdot\vec{E}=\nabla\cdot\left(\frac{1}{4\pi\varepsilon_0}\int_V\frac{\rho(\vec{x}')\vec{r}}{r^3}dV'\right)=\frac{1}{4\pi\varepsilon_0}\int_V\rho(\vec{x}')\,\nabla\cdot\left(\frac{\vec{r}}{r^3}\right)dV'$$

$$=\frac{1}{4\pi\varepsilon_0}\int_V\rho(\vec{x}')\cdot4\pi\,\delta(\vec{r})\,dV'=\frac{1}{\varepsilon_0}\int_V\rho(\vec{x}')\,\delta(\vec{x}-\vec{x}')\,dV'$$

$$=\frac{1}{\varepsilon_0}\cdot\rho(\vec{x})$$

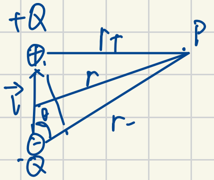
3° E的旋度

$$\vec{E}=\frac{1}{4\pi\varepsilon_0}\int\frac{\rho(\vec{x}')\vec{r}}{r^3}dV'=-\frac{1}{4\pi\varepsilon_0}\int\rho(\vec{x}')\nabla\left(\frac{1}{r}\right)dV'=-\nabla\Big(\underbrace{\frac{1}{4\pi\varepsilon_0}\int\rho(\vec{x}')\cdot\frac{1}{r}dV'}_{\varphi}\Big)$$

$$=-\nabla\varphi$$

$$\nabla\times\vec{E}=\nabla\times(-\nabla\varphi)=0.$$

4° 电偶极子



$$\varphi(P)=\frac{1}{4\pi\varepsilon_0}\frac{Q}{r_+}-\frac{1}{4\pi\varepsilon_0}\frac{Q}{r_-}=\frac{Q}{4\pi\varepsilon_0}\left(\frac{r_--r_+}{r_+r_-}\right)\approx\frac{Q}{4\pi\varepsilon_0}\cdot\frac{l\cos\theta}{r^2}$$

令 $\vec{p}=Q\vec{l}$, $\varphi(P)=\dfrac{\vec{p}\cdot\vec{r}}{4\pi\varepsilon_0 r^2}$

$$\vec{E}_p=-\nabla\varphi_p=-\nabla\left(\frac{\vec{p}\cdot\vec{r}}{4\pi\varepsilon_0r^2}\right)=-\frac{1}{4\pi\varepsilon_0}\cdot\frac{\vec{p}-3(\vec{p}\cdot\vec{r})\hat{r}}{r^3}$$

1.2 电流与磁场

$$dB = \frac{\mu_0}{4\pi}\frac{I\,d\vec{l}\times\hat{r}}{r^2} \qquad dF_{12} = \frac{\mu_0}{4\pi}\frac{(I_2 d\vec{l_2})\times(I_1 d\vec{l_1}\times\vec{r})}{r^3}$$

$$\vec{B} = \frac{\mu_0}{4\pi}\cdot\int\frac{I\,d\vec{l}\times\hat{r}}{r^2} = \frac{\mu_0}{4\pi}\int\frac{\vec{J}\,dV\times\hat{r}}{r^3}$$

$$\boxed{\frac{\partial\rho}{\partial t} + \nabla\cdot\vec{J} = 0}$$

$$\boxed{\nabla\times\left(\frac{\vec{J}}{r}\right) = \underbrace{\nabla\times\vec{J}(x')}_{0.}\,\frac{1}{r} + \vec{J}\times\nabla\left(\frac{1}{r}\right).}$$

1° $\vec{B}$ 的散度

$$\vec{B} = \frac{\mu_0}{4\pi}\int\frac{\vec{J}(x')\times\vec{r}}{r^3}\,dV' = \frac{\mu_0}{4\pi}\cdot\int\vec{J}(x')\times\nabla\left(\frac{1}{r}\right)dV'$$

$$= \frac{\mu_0}{4\pi}\int\nabla\times\left(\frac{\vec{J}(x')}{r}\right)dV' = \nabla\times\underbrace{\int\frac{\mu_0}{4\pi}\cdot\frac{\vec{J}(x')}{r}\,dV'}_{\vec{A}\ \text{磁矢势}.} = \nabla\times\vec{A}$$

$$\nabla\cdot\vec{B} = 0.$$

2° $\vec{B}$ 的旋度

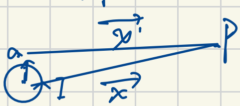
$$\nabla\times\vec{B} = \nabla\times(\nabla\times\vec{A}) = \nabla(\nabla\cdot\vec{A}) - \nabla^2\vec{A}$$

$$\nabla\cdot\vec{A} = \frac{\mu_0}{4\pi}\int\nabla\cdot\left(\frac{\vec{J}(x')}{r}\right)dV' = 0$$

$$\nabla^2\vec{A} = -\mu_0 J(\vec{x})$$

$$\Rightarrow \nabla\times\vec{B} = \mu_0 J(\vec{x})$$

3° 磁偶极子



$$\vec{A} = \frac{\mu_0}{4\pi}\int\frac{\vec{J}(x')}{r}\,dV' = \frac{\mu_0}{4\pi}\int\frac{I\,d\vec{l}}{r} = \frac{\mu_0 I}{4\pi}\int\frac{d\vec{l}}{r}$$

$$约\ \frac{\mu_0 I}{4\pi}\frac{\vec{m}\times\hat{x}}{x^2}$$

$$\vec{m} = I_0\vec{S} \qquad \vec{B} = \nabla\times\vec{A} = -\frac{\mu_0}{4\pi}\frac{\vec{m} - 3(\vec{m}\cdot\hat{r})\hat{r}}{r^3}$$

真空中

	静电场	静磁场
常数	$\frac{1}{\varepsilon_0}$	μ_0
源	ρ	$\vec{J}$
场	$\frac{1}{4\pi\varepsilon_0}\int\frac{\rho\hat{r}}{r^2}dV$	$\frac{\mu_0}{4\pi}\int\frac{\vec{J}\times\hat{r}}{r^2}dV$
散	$\nabla\cdot\vec{E}=\frac{\rho(\vec{x})}{\varepsilon_0}$	$\nabla\cdot\vec{B}=0$
旋	$\nabla\times\vec{E}=0$	$\nabla\times\vec{B}=\mu_0\vec{J}$
势	$E=-\nabla\varphi$	$\vec{B}=\nabla\times\vec{A}$
	$\varphi=\frac{1}{4\pi\varepsilon_0}\int\frac{\rho}{r}dV$	$\vec{A}=\frac{\mu_0}{4\pi}\int\frac{\vec{J}}{r}dV$
极子	$\vec{p}=Q\vec{l}$	$\vec{m}=I\cdot\vec{S}$.
	$\vec{E}=-\frac{1}{4\pi\varepsilon_0}\frac{\vec{p}-3(\vec{p}\cdot\hat{r})\hat{r}}{r^3}$	$\vec{B}=-\frac{\mu_0}{4\pi}\frac{\vec{m}-3(\vec{m}\cdot\hat{r})\hat{r}}{r^3}$

1.3 MAXWELL方程组

法拉第：$\nabla\times\vec{E} = -\frac{\partial B}{\partial t}$

1°

$$\begin{cases} \nabla\cdot\vec{E} = \frac{\rho}{\varepsilon_0} & \to \text{源是电荷} \\ \nabla\times\vec{E} = -\frac{\partial B}{\partial t} & \to \text{磁生电} \\ \nabla\cdot\vec{B} = 0 & \to \text{磁场无源} \\ \nabla\times\vec{B} = \mu_0\vec{J} + \underbrace{\mu_0\varepsilon_0\frac{\partial E}{\partial t}}_{\text{位移电流}} & \to \text{电生磁} \end{cases}$$

$\boxed{\text{真空中}}$

洛伦兹力：$\vec{F} = q\vec{E} + q\vec{v}\times\vec{B}$

电磁力

2° 介质中极化、磁化后的磁/电场

$$\begin{cases} \vec{D} = \varepsilon_0\vec{E} + \vec{P} \to \text{极化矢量} \\ \vec{H} = \frac{1}{\mu_0}\vec{B} - \vec{M} \to \text{磁化矢量} \end{cases}$$

$$\begin{cases} \nabla\cdot\vec{D} = \rho \\ \nabla\times\vec{E} = -\frac{\partial B}{\partial t} \\ \nabla\cdot\vec{B} = 0 \\ \nabla\times\vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} \end{cases}$$

$\boxed{\text{介质中}}$

3° 积分形式

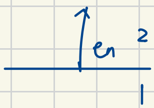
$\int\vec{D}\,d\vec{S} = Q \qquad \Leftarrow \int\varepsilon_0\vec{E}\,d\vec{S} = \int\rho\,dV = Q$

$\oint\vec{E}\cdot d\vec{l} = -\frac{\partial}{\partial t}\int\vec{B}\cdot d\vec{S} \qquad \Leftarrow \qquad \oint\vec{E}\,d\vec{l} = \iint\nabla\times\vec{E}\,d\vec{S} = -\iint\frac{\partial\vec{B}}{\partial t}\,d\vec{S}.$

$\int \vec{B}\, d\vec{S} = 0$

$\oint \vec{H}\, d\vec{l} = I + \frac{d}{dt}\int \vec{D}\cdot d\vec{S}$.

1.4 电磁场的边值关系



$$\begin{cases} \vec{e_n}(\vec{D_2}-\vec{D_1}) = \rho_f \\ \vec{e_n}\times(\vec{E_2}-\vec{E_1}) = 0 \\ \vec{e_n}\cdot(\vec{B_2}-\vec{B_1}) = 0 \\ \vec{e_n}\times(\vec{H_2}-\vec{H_1}) = J_f \end{cases}$$

ρ_f 电荷面密度

$\nabla(\vec{e_n}\times(\vec{E_1}-\vec{E_2}))$

J_f 电流线密度

1.5 能量和能流

能量密度 $w = \frac{1}{2}(\varepsilon_0 E^2 + \frac{1}{\mu_0}B^2)$

能流密度 $\vec{S} = \frac{1}{\mu_0}\vec{E}\times\vec{B}$ $\boxed{\Leftarrow \text{坡印廷矢量}}$

2.1 静电场

$$\begin{cases} \nabla\cdot\vec{E} = \dfrac{\rho}{\varepsilon_0} \\ \nabla\times\vec{E} = 0 \end{cases}$$

无移动电荷
无磁场

泊松方程: $\vec{E} = -\nabla\varphi$ $\qquad$ $\nabla\cdot\vec{E} = \nabla(-\nabla\varphi) = -\nabla^2\varphi = \dfrac{\rho}{\varepsilon_0}$

$$\Longrightarrow \boxed{\nabla^2\varphi = -\frac{\rho}{\varepsilon_0}} \qquad \boxed{\text{当}\rho=0\text{时}: \quad \nabla^2\varphi = 0}$$ ，拉普拉斯方程

边值关系的另一形式

$$\vec{e_n}\cdot(\vec{D_2}-\vec{D_1}) = \rho_f \Longrightarrow \vec{e_n}(-\varepsilon_2\nabla\varphi_2 + \varepsilon_1\nabla\varphi_1) = \rho_f$$

$$\Rightarrow \boxed{\varepsilon_2\frac{\partial\varphi_2}{\partial n} - \varepsilon_1\frac{\partial\varphi_1}{\partial n} = -\rho_f.}$$

$$\vec{e_n}\times(\vec{E_2}-\vec{E_1}) = 0 \Longrightarrow \vec{E_2} = -\nabla\varphi_2, \quad \vec{E_1} = -\nabla\varphi_2$$

$$d\varphi_2 = \frac{\partial\varphi_2}{\partial x}dx + \frac{\partial\varphi_2}{\partial y}dy + \frac{\partial\varphi_2}{\partial z}dz$$

$$= \nabla\varphi_2\cdot(dx, dy, dz)$$

$$= \nabla\varphi_2 d\vec{l} = \boxed{-\vec{E_2}d\vec{l}}$$

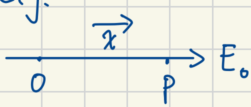
故 $\varphi_2 = \int -\vec{E_2}d\vec{l}$ ， $\varphi_1 = \int -\vec{E_1}d\vec{l}$

$$\boxed{\varphi_2 - \varphi_1 = -\int(\vec{E_2}-\vec{E_1})d\vec{l} = -\iint\nabla\times(\vec{E_2}-\vec{E_1})d\vec{S} = 0.}$$

电场的能量 $W = \frac{1}{2}\int_\infty \varepsilon_0 E^2 dV = \frac{1}{2}\int_\infty \varphi\rho\, dV.$

$$\boxed{W = \Sigma q\varphi}$$

e.g.



$\varphi_p - \varphi_0 = ?$ $\qquad$ $\varphi_p - \varphi_0 = \int_p^0 \vec{E}d\vec{l} = -E_0\cdot x$

$\boxed{\text{电势求法：先算 } \vec{E}\text{，再用 } \varphi = \int \vec{E}\, d\vec{l}}$

$$\varphi_B - \varphi_A = -\int_A^B \vec{E}\, d\vec{l} = \frac{-W_{A\to B}}{q}.$$

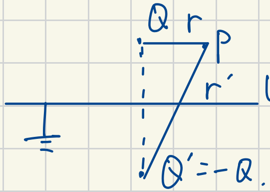
2.2 唯一性定理

$$\nabla^2 \varphi = \frac{-\rho}{\varepsilon_0} \quad / \quad \nabla^2 \varphi = 0$$

$$\begin{cases} \varphi_1 = \varphi_2 \\ \varepsilon_2 \frac{\partial \varphi_2}{\partial n} - \varepsilon_1 \frac{\partial \varphi_1}{\partial n} = -\sigma_f \end{cases}$$

边值条件确一，方程的解唯一

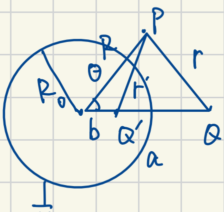
2.3 镜像法



求电势

解：此接地平面等价于 Q 与 Q' 形成的电势面

$$\varphi(P) = \frac{Q}{4\pi\varepsilon_0}\left(\frac{1}{r} - \frac{1}{r'}\right)$$



求电势.

$$\boxed{\varphi(P) = \frac{Q}{4\pi\varepsilon_0 r} - \frac{1}{4\pi\varepsilon_0 r'} \cdot \frac{R_0}{a} Q}$$

Q' 位置如何解得？

$$\begin{cases} \nabla^2 \varphi = \frac{-\rho}{\varepsilon_0} = \frac{-Q\,\delta(x-a)}{\varepsilon_0}. \\ \varphi|_{R=R_0} = 0, \quad \varphi|_{R\to\infty} = 0. \end{cases}$$

$$\varphi = \frac{1}{4\pi\varepsilon_0}\left(\frac{Q}{\sqrt{R^2+a^2-2Ra\cos\theta}} + \frac{Q'}{\sqrt{R^2+b^2-2Rb\cos\theta}}\right)$$

$$R = R_0 \text{时} \Rightarrow \frac{Q^2}{R_0^2+a^2-2R_0 a\cos\theta} = \frac{Q'^2}{R_0^2+b^2-2R_0 b\cos\theta}$$

$$\Rightarrow \boxed{\begin{cases} b = \frac{R_0^2}{a} \\ Q' = -\frac{R_0}{a} Q \end{cases}}$$

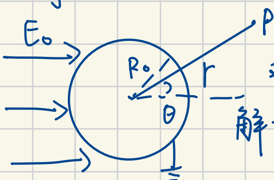
2.4 分离变量法

只用于 $\nabla^2\varphi=0$. 把 $\varphi(x,y,z)$ 拆成 $X(x)Y(y)Z(z)$

$$\varphi=\sum_l \left(A_l r^l+\frac{B_l}{r^{l+1}}\right)\underline{P_l(\cos\theta)}$$ 此为通解

$$\begin{cases} l=0,\ P_l=1 \\ l=1,\ P_l=\cos\theta \\ l=2,\ P_l=(3\cos^2\theta-1)/2 \end{cases}$$

eg1.



求电势分布

解: $\nabla^2\varphi=0$

$\varphi_{R=R_0}=0$, $\varphi_{R\to\infty}=-\vec{E_0}\cdot\vec{r}=-E_0 r\cos\theta$.

$$\varphi=\left(A_l r^l+\frac{B_l}{r^{l+1}}\right)P_l(\cos\theta)$$

$\because \varphi_{r\to\infty}=-E_0 r\cos\theta$ 故 $l=1$ (提 $\cos\theta$)

$\varphi_{r\to\infty}=\left(A_1 r+\frac{B_1}{r^2}\right)\cos\theta$. $=A_1 r\cos\theta=-E_0 r\cos\theta$ $A_1=-E_0$

$\because \varphi_{r\to R_0}=0$ 故 $\varphi_{r\to R_0}=\left(-E_0R_0+\frac{B_1}{R_0^2}\right)\cos\theta=0$

$B_1=E_0R_0^3$.

$$\Rightarrow \varphi=\left(-E_0 r+\frac{E_0R_0^3}{r^2}\right)\cos\theta$$

3.1 静磁场

1° $\begin{cases} \nabla\cdot\vec{B}=0 \\ \nabla\times\vec{H}=\vec{J} \end{cases}$ 无电场 $\quad \vec{B}=\nabla\times\underbrace{\vec{A}}_{\text{矢势}} \quad \vec{B}=\mu\vec{H}$

$$\boxed{\begin{aligned}\nabla\cdot\vec{E}&=\frac{1}{\varepsilon_0}\rho \\ \nabla\times\vec{E}&=0\end{aligned}}$$

$$\nabla\times(\nabla\times\vec{A})=\nabla(\boxed{\underbrace{\nabla\cdot\vec{A}}_{=0}})-\nabla^2\vec{A}=-\nabla^2\vec{A}=\mu\vec{J} \Rightarrow \nabla^2\vec{A}=-\mu\vec{J}$$

$\boxed{\text{泊松方程}\quad \nabla^2 A_i=-\mu_0 J_i}$ V.S $\boxed{\nabla^2\varphi=-\frac{\rho}{\varepsilon_0}}$

磁矢势推导毕-萨

$$\vec{A}=\frac{\mu_0}{4\pi}\int\frac{\vec{J}}{r}dV \qquad \vec{B}=\nabla\times\vec{A}=\frac{\mu_0}{4\pi}\int\nabla\times\frac{\vec{J}}{r}dV=\frac{\mu_0}{4\pi}\int\nabla\frac{1}{r}\times\vec{J}\,dV=\frac{\mu_0}{4\pi}\int\frac{\vec{J}\times\hat{r}}{r^2}dV$$

$$=\frac{\mu_0}{4\pi}\int\frac{I\,d\vec{l}\times\hat{r}}{r^2}$$

$$\varphi=\frac{1}{4\pi\varepsilon_0}\int\frac{\rho}{r}dV$$

$$\nabla\times\left(\frac{\vec{A}}{\mu}\right)=\nabla\times\vec{A}\cdot\frac{1}{\mu}+\underbrace{\vec{A}\times\left(\nabla\frac{1}{\mu}\right)}_{0}$$

2° $\begin{cases} \vec{e_n}\times(\vec{H_2}-\vec{H_1})=\vec{\alpha} \quad \text{电流线密度} \\ \vec{e_n}\cdot(\vec{B_2}-\vec{B_1})=0 \end{cases}$

其矢势形式： $\vec{e_n}\times\left(\frac{1}{\mu_2}\vec{B_2}-\frac{1}{\mu_1}\vec{B_1}\right)=\vec{e_n}\times\left(\frac{1}{\mu_2}\nabla\times\vec{A_2}-\frac{1}{\mu_1}\nabla\times\vec{A_1}\right)$ $\boxed{=\vec{e_n}\times\left(\nabla\times\frac{\vec{A_2}}{\mu_2}-\nabla\times\frac{\vec{A_1}}{\mu_1}\right)=\vec{\alpha}}$

$$\boxed{\varepsilon_2\frac{\partial\varphi_2}{\partial n}-\varepsilon_1\frac{\partial\varphi_1}{\partial n}=\vec{J}}$$

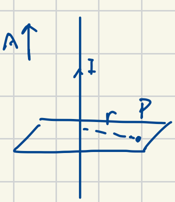
$$\vec{e_n}\cdot(\nabla\times\vec{A_2}-\nabla\times\vec{A_1})=\nabla\cdot(\vec{e_n}\times\vec{A_2}-\vec{e_n}\times\vec{A_1})=0$$

$\Rightarrow \boxed{\vec{A_1}=\vec{A_2}}$ $\qquad \boxed{\varphi_1=\varphi_2}$

3° 静磁场的能量

$$W=\frac{1}{2}\int\vec{A}\cdot\vec{J}\,dV=\frac{1}{2}\int\frac{1}{\mu_0}\vec{B}^2\,dV.$$

$$\boxed{W=\frac{1}{2}\int\rho\varphi\,dV=\frac{1}{2}\int\varepsilon_0\vec{E}^2\,dV}$$



求长直细导线的矢势和 $\vec{B}$

$$\oint\vec{B}\cdot d\vec{l}=B\cdot2\pi r=\mu_0 I$$

$$\vec{B}=\frac{\mu_0 I}{2\pi r}\hat{e}_\varphi \qquad \vec{A}=\frac{\mu_0}{4\pi}\int\frac{\vec{J}}{r}dV=\frac{\mu_0}{4\pi}\int\frac{I}{r}d\vec{l}=\frac{\mu_0 I}{4\pi}\int\frac{dz}{\sqrt{r^2+z^2}}=-\left(\frac{\mu_0 I}{2\pi}\ln\frac{R}{R_0}\right)\hat{z}$$

3.2 磁矢势解法与磁标势

1° 矢势解法

对于二维问题：$\vec{J} = J(x,y)\cdot\hat{z} \Rightarrow \vec{A} = A(x,y)\hat{z}$

$$\nabla^2\vec{A} = -\mu\vec{J} \quad \Rightarrow \quad \nabla^2 A(x,y) = -\mu J(x,y)$$

e.g. 求 $\vec{A}$

$\vec{B} = B_0\hat{x}$

$$\vec{A} = A(x,y)\hat{z}.$$

$$\vec{B} = \nabla\times\vec{A} = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 0 & A(x,y) \end{vmatrix} = \frac{\partial A(x,y)}{\partial y}\vec{i} \qquad \text{而} \ \frac{\partial A(x,y)}{\partial y} = B_0$$

$$A(x,y) = B_0 y + f(x) \qquad \text{取} \ A(x,y) = B_0 y$$

2° 矢势规范自由度

$$A' = A + \nabla f \qquad \nabla\times A' = \nabla\times(A+\nabla f) = \nabla\times A + \nabla\times(\nabla f) = \nabla\times A$$

A' & A 都可以是矢势

3° 磁标势

当 $\nabla\times\vec{H} = \vec{J} = 0 \Rightarrow \vec{H} = -\nabla\varphi_m$ $\qquad \boxed{\nabla\times\vec{E} = 0}$

磁标势 φ_m

$$\nabla\cdot\vec{B} = 0 = \nabla\cdot(\mu\vec{H}) = \mu\nabla(-\nabla\varphi_m) = -\mu\nabla^2\varphi_m$$

$\Rightarrow \nabla^2\varphi_m = 0$ 拉普拉斯方程 (没有 $\vec{J}$)

$\boxed{\nabla^2\varphi = 0 \quad (\text{没有}\ \rho)}$

$$\begin{cases} \vec{e}_n\cdot(\vec{B}_2 - \vec{B}_1) = 0 \\ \vec{e}_n\times(\vec{H}_2 - \vec{H}_1) = \vec{\alpha} = 0 \end{cases} \Rightarrow \begin{cases} \varphi_{m1} = \varphi_{m2} \\ \mu_1\frac{\partial\varphi_{m1}}{\partial n} = \mu_2\frac{\partial\varphi_{m2}}{\partial n} \end{cases}$$

4° 硬铁磁

对铁磁问题：$\vec{B}=\mu_0(\vec{H}+\vec{M})$

$\nabla\cdot\vec{B}=\mu_0\nabla\cdot(\vec{H}+\vec{M})=\nabla\cdot\vec{H}+\nabla\cdot\vec{M}=0 \implies \nabla\cdot\vec{H}=-\nabla^2\varphi_m=-\nabla\cdot\vec{M}$

$$\nabla^2\varphi_m=\nabla\cdot\vec{M}=\frac{-\mu_0\nabla\cdot\vec{M}}{\mu_0}$$

令 $\rho_m=-\mu_0\nabla\cdot\vec{M}$ $\qquad \nabla^2\varphi_m=-\frac{\rho_m}{\mu_0}$ $\qquad \boxed{\nabla^2\varphi=-\frac{\rho}{\varepsilon_0}}$

$$\begin{cases}\varphi_{m1}=\varphi_{m2}\\ \mu_0\frac{\partial\varphi_{m2}}{\partial n}-\mu_0\frac{\partial\varphi_{m1}}{\partial n}=\mu_0(\vec{M_2}-\vec{M_1})\cdot\vec{e}_n\end{cases}$$

4.1 电磁波的传播

1° 波动方程：$\nabla^2 f = \frac{1}{v^2}\frac{\partial^2 f}{\partial t^2}$ $\qquad v$ 相速度

$$\begin{cases} \nabla\cdot\vec{D} = \rho \\ \nabla\times\vec{E} = -\frac{\partial\vec{B}}{\partial t} \\ \nabla\cdot\vec{B} = 0 \\ \nabla\times\vec{H} = \vec{J} + \frac{\partial\vec{D}}{\partial t} \end{cases} \xrightarrow{\rho=0,\ \vec{J}=0} \begin{cases} \nabla\cdot\vec{D} = 0 \\ \nabla\times\vec{E} = -\frac{\partial B}{\partial t} \\ \nabla\cdot\vec{B} = 0 \\ \nabla\times\vec{H} = \frac{\partial\vec{D}}{\partial t} \end{cases}$$

$$\nabla\times(\nabla\times\vec{E}) = \nabla\times\left(-\frac{\partial\vec{B}}{\partial t}\right) = -\frac{\partial}{\partial t}(\nabla\times\vec{B}) = -\frac{\partial}{\partial t}\left(\varepsilon\mu\frac{\partial E}{\partial t}\right) = -\varepsilon\mu\frac{\partial^2 E}{\partial t^2}.$$

$$\nabla\times(\nabla\times\vec{E}) = \nabla(\nabla\cdot\vec{E}) - \nabla^2\vec{E} = -\nabla^2\cdot\vec{E}$$

$$\Rightarrow \nabla^2\cdot\vec{E} = \varepsilon\mu\frac{\partial^2 E}{\partial t^2}$$

$$\boxed{V = \frac{1}{\sqrt{\varepsilon_0\mu_0}} \Rightarrow \text{光速}}$$

2° 波动方程通解

$$\nabla^2\psi - \frac{1}{v^2}\frac{\partial^2\psi}{\partial t^2} = 0 \quad \Rightarrow \quad \psi = e^{i(\vec{k}\cdot\vec{x} - \omega t + \varphi)}$$

ω: 圆频率 $\quad T = \frac{2\pi}{\omega} \quad \vec{k}$ 波矢 $\quad \lambda = \frac{2\pi}{k} \quad V = \frac{\omega}{k}$ 相速度

$\vec{k}$ 波矢 $\downarrow$ 传播方向

$$\frac{1}{\sqrt{\mu_0\varepsilon_0}} = c = \frac{\omega}{k}$$

3° 电磁波特性（横波）

1) $\vec{k}\perp\vec{E}$ $\qquad$ 横波

$\vec{k}\perp\vec{B}$

2) $\vec{E} \perp \vec{B}$

3) $\vec{E}$ 与 $\vec{B}$ 同相位

4) $\frac{E}{B} = c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$

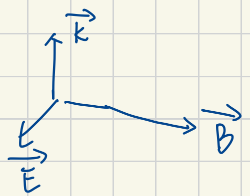
$$\vec{B} = \vec{B_0} e^{\overbrace{i(\vec{k}\cdot\vec{x} - \omega t + \varphi)}^{\text{同相位}}}$$

$$\vec{E} = \frac{1}{\sqrt{\mu_0\varepsilon_0}} \vec{B_0} e^{i(\vec{k}\cdot\vec{x} - \omega t + \varphi)}$$

5) 能量密度 $w = \frac{1}{2}\left(\varepsilon E^2 + \frac{1}{\mu} B^2\right) = \varepsilon E^2 = \frac{1}{\mu} B^2$

6) 能流密度 $\vec{S} = \vec{E} \times \vec{H} = \frac{1}{\mu} \vec{E} \times \vec{B} = \frac{1}{\mu} EB \cdot \hat{k} = \frac{1}{\mu} cB^2 \cdot \hat{k}$

$$= \frac{1}{\mu} \cdot \frac{1}{\sqrt{\mu_0\varepsilon_0}} \cdot \mu w \cdot \hat{k} = \frac{1}{\sqrt{\mu_0\varepsilon_0}} w \cdot \hat{k}$$



4° 电磁波的偏振

$\vec{E} = E_x \hat{x} + E_y \hat{y}$

$$\begin{cases} \phi_x = \phi_y & : \text{线偏振} \\ |\phi_x - \phi_y| = \frac{\pi}{2} & : \text{圆偏振} \\ \phi_x - \phi_y = \theta & : \text{椭圆偏振} \end{cases}$$

5° 电磁波的折射与反射

布儒斯特角：反射光只有偏振光

全反射（无折射）：$n_{入射} > n_{折射}$

5.1 电磁波的辐射

1°
$$\begin{cases} \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \\ \nabla \cdot \vec{B} = 0 \end{cases}$$
有磁场的电场

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = -\frac{\partial}{\partial t}(\nabla \times \vec{A}) = -\nabla \times \frac{\partial \vec{A}}{\partial t} \Rightarrow \nabla \times \left(\vec{E} + \frac{\partial \vec{A}}{\partial t}\right) = 0$$

$$\Rightarrow \vec{E} + \frac{\partial \vec{A}}{\partial t} = -\nabla \cdot \varphi \qquad \Rightarrow \vec{E} = -\nabla \varphi - \frac{\partial \vec{A}}{\partial t}$$

2°
$$\begin{cases} \text{库仑规范} \quad \nabla \cdot \vec{A} = 0 & \text{无辐射} \\ \boxed{\text{洛伦兹规范} \quad \nabla \cdot \vec{A} + \frac{1}{c^2}\frac{\partial \varphi}{\partial t} = 0} & \text{有辐射} \end{cases}$$

$$\nabla \times \vec{B} = \nabla \times (\nabla \times \vec{A}) = \mu_0 \vec{J} + \varepsilon_0 \mu_0 \frac{\partial}{\partial t}\left(-\nabla \varphi - \frac{\partial \vec{A}}{\partial t}\right). \qquad ①$$

$$\nabla \cdot \vec{E} = \nabla \cdot \left(-\nabla \varphi - \frac{\partial \vec{A}}{\partial t}\right) = -\nabla^2 \varphi - \frac{\partial}{\partial t}\nabla \cdot \vec{A} = \frac{\rho}{\varepsilon_0} \qquad ②$$

$\frac{1}{c^2} = \mu_0 \varepsilon_0$，代入

$$\begin{cases} \nabla^2 \vec{A} - \frac{1}{c^2}\frac{\partial^2 \vec{A}}{\partial t^2} - \nabla\left(\nabla \cdot \vec{A} + \frac{1}{c^2}\frac{\partial \varphi}{\partial t}\right) = -\mu_0 \vec{J} & ①' \\ \underbrace{\nabla^2 \varphi + \frac{\partial}{\partial t}\nabla \cdot \vec{A} = -\frac{\rho}{\varepsilon_0}} & ②' \end{cases}$$

$$\Downarrow \text{洛伦兹规范}$$

$$\begin{cases} \nabla^2 \varphi - \frac{1}{c^2}\frac{\partial^2 \varphi}{\partial t^2} = -\frac{\rho}{\varepsilon_0} \\ \nabla^2 \vec{A} - \frac{1}{c^2}\frac{\partial^2 \vec{A}}{\partial t^2} = -\mu_0 \vec{J} \end{cases}$$
达朗贝尔方程 $(\vec{A}, \varphi)$

$$\boxed{\begin{matrix} \nabla^2 \varphi = -\frac{\rho}{\varepsilon_0} \\ \nabla^2 \vec{A} = -\mu_0 \vec{J} \end{matrix}}$$

5.2 推迟势

$$\nabla^2\varphi-\frac{1}{c^2}\frac{\partial^2\varphi}{\partial t^2}=-\frac{\rho}{\varepsilon_0}$$

源的位置 推迟的体现

通解 $\varphi(\vec{x},t)=\dfrac{Q(\vec{x}',t-\frac{r}{c})}{4\pi\varepsilon_0 r}=\displaystyle\int_V\frac{\rho(\vec{x}',\boxed{t-\frac{r}{c}})}{4\pi\varepsilon_0 r}dV'$

同理: $\vec{A}(\vec{x},t)=\dfrac{\mu_0}{4\pi}\displaystyle\int_V\frac{\vec{J}(x',t-\frac{r}{c})}{r}dV$.

5.3 电偶极辐射

6.1 狭义相对论

1. ① 相对性原理：所有惯性参考系等价

② 光速不变原理：波动方程不变

2° 洛伦兹变换

$$t' = \frac{t - \frac{u}{c^2}x}{\sqrt{1-u^2/c^2}} \qquad x' = \frac{x-ut}{\sqrt{1-u^2/c^2}}$$

$$\Delta t = \frac{\Delta \tau}{\sqrt{1-u^2/c^2}} \qquad l = l_0\sqrt{1-u^2/c^2} = \frac{l_0}{\gamma}$$

$$v_x' = \frac{v_x - u}{1-\frac{uv_x}{c^2}}$$ 参考系速度

$$v_y' = \frac{v_y\sqrt{1-u^2/c^2}}{1-\frac{uv}{c^2}}$$

$$v_z' = \frac{v_z\sqrt{1-u^2/c^2}}{1-\frac{uv}{c^2}}$$